

Reproducing published meta-analyses by re-extracting the data of the original trials with local language models

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Abstract

Meta-analyses guide clinical decisions, but their evidence tables are extracted by hand and carry errors that sometimes change conclusions. Language models can re-extract the data from the original trials, but the existing tests use cloud services, do not distinguish reading from recitation (repeating published values from memory) and treat the published review as an infallible answer key. In this pre-registered study, six open-weight language models, on a single consumer computer and without sending any text outside it, re-extracted the 29 randomized clinical trials of three published meta-analyses: goal-directed fluid therapy in major surgery (morbidity and mortality), low-carbohydrate diet in type 2 diabetes (glycated hemoglobin) and methylene blue in adult shock (mortality at 28–30 days). For each trial, each model filled in a sheet with the values the meta-analysis uses (patients per arm, events, means and standard deviations); each value is a cell. The models read copies of the trials in which the key numbers had been altered, so that recitation would be detectable, and each cell was compared with the value verified directly in the trial, with a literal quotation; the same comparison was applied to the tables the reviews published, to see what the original reviews got right. The reviews got 82%, 100% and 91% of the cells right; in the third, two transcription errors (survivors entered as deaths in one trial; the deaths of two arms over the denominator of one in another), once corrected, move the odds ratio of death with methylene blue from 0.73 [0.40; 1.36] to 0.48 [0.34; 0.69], a benefit where the review concluded there was none. The models got 69% to 90% of the cells right in the first meta-analysis, 38% to 73% in the second and 57% to 100% in the third, depending on the model and the sheet; all reproduced the pooled risk ratios of the first, the mean difference of the second separated the models, none reproduced the errors of the third, and two 14-billion-parameter models (`qwen3:14b` and `deepseek-r1:14b`) got all 32 cells right and arrived exactly at the corrected estimate. Of the 619 cells in which a model did not match the trial, 346 were right and merely written in another form, 147 were omissions, 105 reading errors and 21 errors of the grading program; none was recitation. Trial-faithful extraction by a local model reproduces published meta-analyses and exposes errors that change conclusions; checking should concentrate on the labels (which population was counted, what kind of dispersion was printed), where humans and models both err.

1 Introduction

Systematic reviews and meta-analyses are the reference standard of clinical evidence synthesis, with mature reporting standards [Page et al., 2021], and they depend on human labor: a registered review takes 67 weeks on average from registration to publication [Borah et al., 2017], most of it spent extracting, checking and pooling the trial data. That hand extraction errs: 17 of 27 meta-analyses using standardized mean differences contained an extraction error in at

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least one audited trial, some large enough to reverse a finding [Gøtzsche et al., 2007]; extraction and reporting errors were found in most Cochrane reviews of an earlier audit [Jones et al., 2005]; independent observers extracting the same continuous outcomes produced meta-analyses that disagreed [Tendal et al., 2009], in part because trial reports offer several candidate values for the same quantity [Tendal et al., 2011]; and the evidence base of the extraction standards remains weak [Mathes et al., 2017].

Large language models are entering this workflow. Feasibility studies report promising but uneven accuracy in extracting trial data [Schmidt et al., 2024, Khraisha et al., 2024, Konet et al., 2024]; assisted extraction proved noninferior to human-only extraction in a live review [Lee et al., 2026]; screening assistance is approaching routine use [Xie et al., 2026]; and end-to-end platforms now report complete reviews in hours [Bakker et al., 2026], with automated drafts rated above a human-authored review by blinded specialists [McLaughlin et al., 2026]. The evidence-synthesis organizations condition that use on demonstrated rigor, human oversight and transparent reporting [Flemyng et al., 2025], and a reporting guideline for studies that use these models already exists [Gallifant et al., 2025]. Three problems run through the evaluation literature. Most tests use commercial cloud models, while clinical text asks for the opposite: open-weight models, which are downloaded and run on one’s own machine, keep the material in-house and have already shown extraction close to expert annotation [Wiest et al., 2024, Spaanderman et al., 2025]. A test that scores agreement with a published review does not distinguish a model that *read* the trial from one that *remembers* the review, since reviews and trials are in the training texts and that contamination inflates apparent competence [Carlini et al., 2023, Xu et al., 2024, Golchin and Surdeanu, 2024]. And the published review is usually treated as an infallible answer key, so that a test does not notice when the published value is wrong and misgrades precisely the best readers.

However, no evaluation, to our knowledge, combines the three controls — local models, a reading-versus-recitation proof and an answer key verified in the trials, able to overrule the review — on more than one kind of effect measure. This study measured six open-weight models, on a single consumer computer, as extractors for three published meta-analyses, and asked whether trial-faithful extraction reproduces the published pooled estimates and, where the published data are wrong, produces the corrected estimate instead.

2 Methods

2.1 Design and registration

The protocol — question, models, meta-analyses, sheets, hypotheses, instruments, tolerances and compute budget — was registered on 2026-09-08 against a fixed version of the public repository, reviewed before registration by five independent reviews (77 findings raised, 43 applied), and not amended afterwards. Texts, alterations, answer keys, pooling programs and graders were built and frozen before the first model call; the models ran on 9 and 10 September 2026. Six hypotheses were pre-registered (Table 6). The study analyzed only published aggregate reports of randomized trials and three published meta-analyses; it involved no human participants and required no ethics approval. All materials are public [Barra, 2026].

2.2 The three meta-analyses and their trials

Three recent meta-analyses, called anchors here, served as targets (Table 1). The first compares goal-directed fluid therapy with conventional fluid therapy in major surgery [Ashraf et al., 2026]: 14 randomized trials, morbidity (4 trials) and mortality (2) outcomes, as risk ratios. The second compares low-carbohydrate diets with control diets in type 2 diabetes [Khraise, 2026]: 7 trials, the outcome being the change in glycated hemoglobin (HbA1c, %), as a mean difference. The

third compares methylene blue with placebo or usual care in adult shock [Rodríguez-Lima et al., 2026]: 8 trials, the outcome being mortality at 28–30 days, as an odds ratio. The 29 trials formed the three corpora (the collections of texts the models read) as plain text: article body and tables converted to text, figures not included. For each trial, a model fills in a sheet with the values the meta-analysis uses — patients per arm, events, means and standard deviations; each extracted value is a cell, and Table 1 gives the number of cells per sheet.

Table 1: The three anchor meta-analyses and their answer keys.

	Anchor 1	Anchor 2	Anchor 3
Clinical question	goal-directed vs conventional fluid therapy, major surgery	low-carbohydrate diet vs control, type 2 diabetes	methylene blue vs placebo or usual care, adult shock
Trials	14	7	8
Text format	8 PMC XML; 6 PDF to text	6 XML; 1 PDF to text	4 XML; 3 PDF to text; 1 PDF in Spanish
Outcome and measure	morbidity (4 trials) and mortality (2), risk ratio	change in HbA1c, % (7 trials), mean difference	mortality at 28–30 days (8 trials), odds ratio
Published	morbidity 0.778 [0.567; 1.068] mortality 1.021 [0.446; 2.337]	−0.24 [−0.32; −0.16]	0.73 [0.40; 1.36]
Verified in the trials (same method)	morbidity 0.787 [0.580; 1.068] mortality 1.023 [0.446; 2.346]	−0.24 [−0.32; −0.16]	0.484 [0.319; 0.735]
Cells per sheet	124	49	32
Confirmed discrepancies	11 (record of 17 entries)	0	2 (3 cells; 2 of 8 trials)
Reading proof	32 altered pairs	17 altered pairs	denominators of 10 of 17 arms (4.5 of 8 trials)

2.3 Verification of the reviews against the trials: the two-layer answer key

Each corpus was graded against a two-layer answer key. Layer 1 holds the cells the review published; layer 2 holds each cell verified against the full text of the trial, with a literal quotation and a verdict, and is the grading ruler: where the two layers disagree, the trial decides and the disagreement is recorded as an erratum of the review. On anchor 1 that verification had produced a quotation-backed record of 11 confirmed discrepancies in the published review (swapped arms, a mislabelled outcome row, counts of analyzed patients presented as randomized). On anchor 2 the published cells reproduced without problem. On anchor 3 the review does not print the per-trial cells in its text: they exist only in its forest plot, the figure that lists each trial with its estimate and, below, the pooled estimate; layer 1 was read from that image and verified by reconstruction (the 16 cells read predict the 48 other values printed in the figure, and all reproduce), and layer 2 found two trials wrong. For Shaker 2025 the review records 15/30 in the methylene blue arm; the trial has three arms (placebo 14/30; methylene blue 9/30 and 6/30), so the published numerator sums the two active arms while the denominator keeps one, and layer 2 holds 15/60 against 14/30, by the Cochrane handbook’s rule for combining arms [Higgins et al., 2023]. For Aguilar 2016 the review records 24/30 and 19/30; the trial reports mortality of 20.0% and 36.6% in arms of 30, that is, 6 and 11 deaths, so the published cells are the survivors, and layer 2 holds 6/30 against 11/30. Three internal inconsistencies of the review were also fixed before the runs: its primary analysis is described in three incompatible ways (the forest plot governs, being the only description with cells); the pooling method is not named, and the printed result reproduces only under one specific random-effects method (Paule–Mandel with the Hartung–Knapp adjustment [Veroniki et al., 2016, IntHout

et al., 2014]); and Aguilar contributes a 21-day value to an analysis labelled 28–30 days (used as the review used it, with its exclusion run as a sensitivity analysis). No verdict rests on the published 0.73. Besides the models, the extractions published by the reviews themselves were assessed in the same way, to see what the original reviews got right: each value of the published table was compared, by the same comparison program, with the value verified in the trial (the *human row* of each anchor). The verification of the trials was done before the runs by the author with the cloud AI assistant, with the literal quotation recorded before the verdict; for example, the composite outcome of Diaper 2021 checks (113, 57.7%, in both), and the arm sizes of Yoon 2023 do not (the review published 36 and 39; the trial says 39 and 36).

2.4 The reading-versus-recitation proof

Before any model read a trial, a deterministic program altered, in the trial’s text, numbers the meta-analysis depends on (anchors 1 and 2: 32 and 17 original-value–altered-value pairs). A model that returns the altered value read the text in front of it; one that returns the original published value is reciting from memory. The models saw only the altered texts; the original↔altered list stayed sealed with the graders and was applied after extraction, returning each altered value to its original for the comparison with the answer key (the reversal lens). The cells of anchor 3 are small integer counts, which cannot be altered without coinciding with other integers in the text, so the proof was placed on the per-arm denominators of 10 of 17 arms, with the dependent percentages and totals recalculated to keep the text coherent; the event counts carry no proof, three trials carry none, and the coverage — four and a half trials of eight — accompanies every anchor-3 result. An automatic screen flagged as a recitation candidate any cell that contained an original value while the altered value was absent; every candidate was adjudicated against the altered text. One further trap was recorded in advance: a sheet that returns the review’s 24/30 for Aguilar reproduced the review, not the trial.

2.5 Models, computer and extraction sheets

Six open-weight models ran as peers: `gemma4:12b`, `qwen3:14b`, `llama3.1:8b`, `qwen3.5:9b` and `deepseek-r1:14b`, all in the compressed Q4_K_M version (about 4.5 bits per weight), and a sixth, `smtek/Qwen3.8-27B:Q2_K_XL`, called `qwen3.8:27b` in the tables and text, with 27.3 billion parameters in a mixed compression measured in the weight file itself as 3.13 bits per weight, because below four bits quality depends on the compression scheme [Dettmers and Zettlemoyer, 2023]. All calls ran on a single consumer mini-computer (AMD Ryzen 7, integrated Radeon 780M graphics, 32 GB of RAM, no dedicated card), served by Ollama with a context window of 24,576 tokens (the maximum text received at once), reasoning mode off, default generation settings and no fixed seed, one model loaded at a time; no cloud service was called at any evaluated step. Two frozen extraction sheets, in English, ran on every anchor — the sheet is the form the model fills in for each trial: v1 asks for each cell a value and the section or table it came from; v2 additionally requires a literal quotation from the trial for each cell (output limits of 4,000 and 8,000 tokens). The anchor-1 sheet covers descriptive and outcome fields; the anchor-2 sheet, the change in HbA1c, its dispersion and the arm sizes; the anchor-3 sheet asks, for each arm of the trial, the event count, the denominator, the reported percentage and the outcome timepoint, with no instruction about the sign (deaths, not survivors) and without reducing three-arm trials to two, a reduction done afterwards by a frozen program.

2.6 Grading, pooling and adjudication of the discordant cells

Phase 1 ran $6 \times (14 + 7 + 8) \times 2 \times 2 = 696$ calls: two replicates (independent runs) per sheet, over the altered texts, by an execution program that accepted only a complete and readable output, recording tokens, stop reason and duration per call. Phase 2 reversed the alteration in each

sheet and compared each cell with layer 2: on anchors 1 and 2, a frozen numeric comparator that accepts differences of format but not of value (124 and 49 eligible cells per sheet, those with a verified value containing digits); on anchor 3, exact integer comparison (32 cells: the event count and the denominator of 16 arms); agreement between replicates used the same comparator. Phase 3 pooled the cells of each sheet with frozen programs and no model in the loop, by the DerSimonian–Laird random-effects method [DerSimonian and Laird, 1986]: risk ratios on anchor 1 over the review’s own per-outcome lists; mean differences on anchor 2 after fixed conversion rules (a declared change used directly; standard errors and confidence intervals converted to standard deviations; baseline–final pairs combined under the $r = 0.5$ rule [Higgins et al., 2023]); odds ratios on anchor 3 after the reduction of three arms to two. On anchor 3 the review’s own method (Paule–Mandel with Hartung–Knapp) was run only by the comparator, to measure the review. Tolerances were ± 0.01 on ratios and interval bounds and ± 0.1 on mean differences. Two automatic provenance checks (called nets) ran over the v2 cells as an exploratory arm: one flags a quotation not found in the trial, the other a value not found in its quotation. On 2026-09-12, after the first version of this record, two defects of this grader were found and fixed with regression tests: the dispersion-type field of the v2 sheet was read as text, so that confidence intervals correctly transcribed by the models were counted as omissions and standard errors as standard deviations, and the interval parser lost the Unicode minus sign; anchor 2 was re-graded, and the numbers reported here replace the earlier ones, which remain in the repository’s history.

Phase 4 adjudicated every discordant cell, one by one, with the trial’s quotation before the verdict. Named automatic rules classified each cell by mechanism (omission; faithful to the trial, written in another form or counting another population; reading slip; error of the grader itself); the cells no rule decided were decided by the reader — the AI assistant, under delegation from the author, every verdict reversible — and all 85 of the reader’s verdicts plus the 39 cells that put the answer key itself on trial were re-read by three independent reviewers, instructed to refute them with the trials open (ten contested, nine corrections accepted after re-verification). The final score adds to the automatic one only the cells in which the model was right and the grader wrong; values written in another form, another population counted and summaries remain discordant. The complete cell-by-cell record, with the quotation and the rule or reader that decided each one, is in the repository.

2.7 Software and analysis

All computation used Python 3.12 and the repository’s validated functions (risk ratio, mean difference and odds ratio with intervals; standard-deviation conversions; DerSimonian–Laird and Paule–Mandel/Hartung–Knapp pooling); no external statistical package was used. Replicates measure stability, not significance; with one corpus per anchor, comparisons between models are descriptive, and the pre-registered hypotheses were read against their fixed criteria.

3 Results

3.1 Execution

All 696 calls completed and passed the integrity check, with no refusal and no truncation, in 39.7 h of summed call time (3.42 min per call; 7.52 million input tokens and 0.78 million output; largest input text 17,898 of 24,576 tokens). The recitation screen raised 13 candidate cells, all on anchor 1, and all 13 resolved as reconstruction from values the alteration program had left in the text (a surviving percentage beside an altered count, a repeated sentence that survived, surviving counts under an altered percentage); no attributable recitation occurred, and the anchor-3 trap (24/30) appeared on no sheet.

3.2 What the original reviews got right

Compared with the data verified in the trials, the extractions published by the reviews got 102 of 124 cells (82%) right in the first meta-analysis, 49 of 49 in the second and 29 of 32 (91%) in the third. In the first, the 22 wrong cells include the swapped arms of Yoon 2023 and the times in hours of Sun 2023 that the trial does not print (it publishes medians in days); 12 of them are the discrepancies confirmed in the verification. In the third, the three wrong cells are the two transcription errors. By the review’s own pooling method, anchor 3’s published estimate changes with each correction (Table 2): from 0.735 [0.397; 1.361] as published to 0.604 [0.325; 1.122] with the Shaker cell corrected, 0.592 [0.385; 0.912] with the Aguilar cell corrected, 0.484 [0.342; 0.685] with both, and 0.493 [0.326; 0.744] with both corrected and Aguilar excluded. The published conclusion — that current randomized evidence does not demonstrate a mortality benefit — survives neither correction nor the simple removal of the disputed trial. On anchors 1 and 2 the discrepancies change neither the direction nor the significance of the published estimates (Table 1).

Table 2: Anchor 3: the odds ratio of death with methylene blue, by the review’s own method, as each transcription error is corrected.

Scenario	OR [95% CI]	Significant effect
As published	0.735 [0.397; 1.361]	no
Shaker 2025 corrected (15/60, not 15/30)	0.604 [0.325; 1.122]	no
Aguilar 2016 corrected (6/30 vs 11/30, not 24/30 vs 19/30)	0.592 [0.385; 0.912]	yes
Both corrected	0.484 [0.342; 0.685]	yes
Both corrected and Aguilar excluded (7 trials)	0.493 [0.326; 0.744]	yes

3.3 Agreement of the models with the trials

Cell agreement with the trial-verified answer key (Table 3) ranged, by model and sheet, from 85/124 to 111/124 on anchor 1, from 16/42 to 36/49 on anchor 2 and from 16/28 to 32/32 on anchor 3; **qwen3.8:27b** held the highest score on anchors 1 and 2 under both sheets, and two 14-billion-parameter models, **qwen3:14b** and **deepseek-r1:14b**, scored 32 of 32 on anchor 3 under v1. On anchor 1 the published review sits below the best readers and above the weakest; on anchor 3, below the two best. Agreement between replicates ranged from 75.9% to 98.3% (anchor 1), from 65.3% to 100% (anchor 2) and from 62.5% to 100% (anchor 3). Under v2 every model scored the same or less on anchor 1 (mean -6.8 cells, range -1 to -20), and three v2 sheets on anchor 3 were refused by the strict JSON reader (the program that opens the model’s answer and rejects invalid characters) because the mandatory quotation carried control characters inherited from text extracted from PDF; a tolerant reader, run as a declared sensitivity, recovered them to 28/32, 20/32 and 30/32.

3.4 Pooled estimates rebuilt by the models

From the cells of each sheet, the frozen programs built anchor 1’s pooled morbidity risk ratio between 0.738 and 0.788 against the verified 0.787, nine of twelve pairs within 0.01 and eleven within 0.02, every interval crossing 1 as published (Table 4); pooled mortality stayed between 0.980 and 1.078 in eleven pairs, and one pair (**qwen3.8:27b**, v1) fell to 0.400 [0.018; 9.121] because, in the de Waal 2021 trial, the model wrote 31 as the size of the control arm, a stray number next to the flowchart’s title, in place of the 259 patients; with the wrong denominator, control mortality was inflated and the risk ratio collapsed. On anchor 2, six of twelve pairs fell within the pre-registered ± 0.1 around the verified -0.24 : **gemma4:12b** v1 at -0.27 $[-0.38; -0.17]$ and v2 at -0.24 $[-0.32; -0.17]$, **qwen3.8:27b** at -0.25 under both sheets, all four

Table 3: Cells agreeing with the trial-verified answer key, by model, sheet and anchor, after the adjudication of the discordant cells. First row: the published review’s own cells on the same ruler. †v2 sheets refused by the strict JSON reader; reduced denominator.

Extraction	Anchor 1 (124 cells)		Anchor 2 (49)		Anchor 3 (32)	
	v1	v2	v1	v2	v1	v2
Published (human)	102		49		29	
gemma4:12b	109	106	33	34	23	23
qwen3:14b	105	85	24	22	32	24/28†
llama3.1:8b	92	86	24	16/42	26	23
qwen3.5:9b	95	91	30	30	28	16/28†
deepseek-r1:14b	109	102	25	24	32	22/24†
qwen3.8:27b	111	110	35	36	28	28

over the seven trials, and **qwen3:14b** v2 (-0.27) and **deepseek-r1:14b** v2 (-0.22) over five; the other six pairs stayed between -0.42 and -0.68 , with five to seven trials pooled. The v2 numbers on this anchor are those re-graded after the grader correction described in Section 2.6; before it, confidence intervals correctly transcribed by the models were counted as omissions, the v2 pools had 2 to 4 trials and the sheet appeared to starve them. In eleven anchor-2 cells the model was right and the reversal lens could not undo what it had read correctly (Section 3.6); with those cells at the right value, seven estimates move toward the trials (**qwen3:14b** v1 from -0.42 to -0.32 , **deepseek-r1:14b** v1 from -0.46 to -0.27 , **qwen3.5:9b** v2 from -0.68 to -0.31 , the others by 0.02 to 0.25), a sensitivity analysis recorded beside the primary.

Table 4: Pooled estimates built by the frozen programs from the cells of each sheet, anchors 1 and 2 (estimate [95% CI]; anchor 2 with the number of trials pooled). RR: risk ratio; MD: mean difference. ‡one stray denominator.

Extraction	Sheet	A1 morbidity RR	A1 mortality RR	A2 HbA1c MD (trials)
Published (human)	—	0.778 [0.567; 1.068]	1.021 [0.446; 2.337]	-0.24 [-0.32; -0.16] (7)
Layer 2 (source)	—	0.787 [0.580; 1.068]	1.023 [0.446; 2.346]	-0.24 [-0.32; -0.16] (7)
gemma4:12b	v1	0.779 [0.569; 1.065]	1.019 [0.445; 2.336]	-0.27 [-0.38; -0.17] (7)
	v2	0.779 [0.569; 1.065]	1.078 [0.470; 2.471]	-0.24 [-0.32; -0.17] (7)
qwen3:14b	v1	0.779 [0.569; 1.065]	1.021 [0.446; 2.337]	-0.42 [-0.91; 0.06] (7)
	v2	0.779 [0.569; 1.065]	1.019 [0.445; 2.336]	-0.27 [-0.45; -0.09] (5)
llama3.1:8b	v1	0.738 [0.510; 1.070]	1.023 [0.447; 2.344]	-0.63 [-1.02; -0.25] (5)
	v2	0.767 [0.552; 1.066]	1.023 [0.447; 2.344]	-0.51 [-0.89; -0.12] (6)
qwen3.5:9b	v1	0.779 [0.569; 1.065]	1.021 [0.446; 2.337]	-0.63 [-1.07; -0.19] (5)
	v2	0.779 [0.569; 1.065]	1.019 [0.445; 2.336]	-0.68 [-1.22; -0.14] (5)
deepseek-r1:14b	v1	0.788 [0.579; 1.072]	0.980 [0.428; 2.246]	-0.46 [-0.66; -0.25] (5)
	v2	0.769 [0.558; 1.061]	1.021 [0.446; 2.337]	-0.22 [-0.57; 0.13] (5)
qwen3.8:27b	v1	0.787 [0.580; 1.068]	0.400 [0.018; 9.121]‡	-0.25 [-0.37; -0.12] (7)
	v2	0.779 [0.569; 1.065]	1.023 [0.446; 2.346]	-0.25 [-0.37; -0.13] (7)

On anchor 3 no sheet of any model reproduced either transcription error: zero cells agreed with the published 24/30 or 15/30 against the trials (Table 5). Under v1, four of six models read Aguilar as 6/30 from the Spanish-language trial, and the other two returned no mortality entry for that trial on either sheet (counted as omissions); six of six enumerated Shaker’s three arms, so that the two-arm reduction yielded 15/60; four of six got both cells right. **qwen3:14b** and **deepseek-r1:14b** got 32 of 32 cells right and pooled, by the campaign’s method, exactly to the trial-verified constant, OR 0.484 [0.319; 0.735], and by the review’s own method to 0.484 [0.342; 0.685], where the review published 0.73 [0.40; 1.36]. Every pair’s estimate stayed between 0.418 and 0.548 with the upper bound below 1; by the review’s method one upper bound reached 1.002 (**gemma4:12b** v1, six trials). Of anchor 3’s 63 discordant cells, 56 were

omissions — five sheets wrote nothing for the Spanish-language trial and four for the trial that publishes no per-arm count — and the seven reading slips were two survivor counts reported as deaths, two percentages written in place of the denominator and three partial cause-specific counts.

Table 5: Anchor 3: the two deciding cells (the methylene blue arm of Aguilar 2016 and of Shaker 2025), the score on 32 cells and the pooled odds ratio by the campaign’s method (DerSimonian–Laird, trials pooled in parentheses) and by the review’s own method (Paule–Mandel with Hartung–Knapp). OR: odds ratio; MB: methylene blue; NR: no mortality entry returned. †as in Table 3.

Extraction	Sheet	Aguilar MB	Shaker MB	Cells	OR, DL (trials)	OR, PM–HK
Published (human)	—	24/30	15/30	29/32	0.737 [0.449; 1.208] (8)	0.73 [0.40; 1.36]
Layer 2 (source)	—	6/30	15/60	32/32	0.484 [0.319; 0.735] (8)	0.484 [0.342; 0.685]
gemma4:12b	v1	NR	15/60	23/32	0.519 [0.321; 0.839] (6)	0.520 [0.270; 1.002]
	v2	NR	15/60	23/32	0.418 [0.246; 0.711] (6)	0.418 [0.211; 0.828]
qwen3:14b	v1	6/30	15/60	32/32	0.484 [0.319; 0.735] (8)	0.484 [0.342; 0.685]
	v2	6/30	15/60	24/28†	0.481 [0.311; 0.745] (6)	0.481 [0.393; 0.589]
llama3.1:8b	v1	6/30	15/60	26/32	0.496 [0.318; 0.773] (7)	0.496 [0.319; 0.770]
	v2	6/30	NR	23/32	0.517 [0.322; 0.832] (6)	0.517 [0.348; 0.770]
qwen3.5:9b	v1	6/30	15/60	28/32	0.460 [0.299; 0.708] (7)	0.460 [0.325; 0.651]
	v2	NR	15/60	16/28†	0.450 [0.267; 0.758] (4)	0.450 [0.212; 0.951]
deepseek-r1:14b	v1	6/30	15/60	32/32	0.484 [0.319; 0.735] (8)	0.484 [0.342; 0.685]
	v2	6/30	15/60	22/24†	0.548 [0.362; 0.831] (6)	0.548 [0.388; 0.775]
qwen3.8:27b	v1	NR	15/60	28/32	0.493 [0.315; 0.770] (7)	0.493 [0.326; 0.744]
	v2	NR	15/60	28/32	0.493 [0.315; 0.770] (7)	0.493 [0.326; 0.744]

3.5 The pre-registered hypotheses

Table 6 lists the six hypotheses (H12.1 to H12.6) and their outcomes. The mandatory-quotation sheet (v2) cost accuracy on anchor 1 and brought the estimates closer to the trials only on anchor 2 (closer in one model on anchor 1, four on anchor 2 and none on anchor 3, against the criterion of four or more on two anchors). The provenance nets flagged 19–39% of the eligible anchor-3 cells on the quotation-exists check and 7–43% on the value-in-quotation check, within the ranges of anchors 1 and 2 (10–57% and 7–45%). The context control fell one cell outside its band on anchor 1 and inside it on anchor 2.

Table 6: Pre-registered hypotheses and outcomes.

	Criterion, fixed before the runs	Result	Verdict
H12.1	every anchor-3 sheet is readable and its pooled estimate is consistent with its own cells (± 0.01)	estimates consistent in 12 of 12 pairs; three v2 sheets refused by the strict JSON reader	9 of 12 (12 of 12 with the tolerant reader)
H12.2	≥ 1 of 6 models writes Aguilar 6/30 (not 24/30) and reduces Shaker to 15/60	v1: Aguilar 6/30 in 4 of 6, Shaker 15/60 in 6 of 6, both in 4; errors reproduced by 0	met
H12.3	v2 costs ≥ -4 cells of 124 on anchor 1 and pools closer to the trials in ≥ 4 of 6 models on ≥ 2 anchors	mean -6.8 cells (range -1 to -20); closer on 1 of 3 anchors	fails both
H12.4	(exploratory) the provenance nets over-fire on anchor 3, whose data live in tables	19–39% and 7–43% of anchor-3 cells vs 10–57% and 7–45% on the others	no excess
H12.5	context control: gemma4:12b on anchor 1 v1 within 103 ± 4 cells and anchor-2 estimate within ± 0.05 of -0.27	109/124 (108 by the automatic rule); estimate -0.27	one cell outside
H12.6	bounded null: no model above 50% of anchor 3’s 32 cells	11 of 12 pairs above 50%	the null falls

3.6 The 619 discordant cells

The adjudication classified the 619 cells that failed the comparison with the trials (297 on anchor 1, 259 on anchor 2, 63 on anchor 3; Table 7). Three hundred and forty-six were faithful to the trial: 165 counted another population (the analyzed patients where the key holds the randomized ones, or the reverse), 78 summarized the type of surgery into a category label instead of listing the surgeries, 54 wrote the percentage in place of the count, 33 carried standard deviations derived by the key’s own rule from the analyzed n the model had read, 6 used another unit (mmol/mol), 6 were cells for which the trial itself prints two contradictory counts, 3 rewrote only part of the datum and 1 spelled out “not reported”. One hundred and forty-seven were omissions. One hundred and five were reading slips, 73 of them on the continuous anchor: 58 values taken from another row or scope (a between-group difference written as one arm’s change, another visit, the analyzed n of another outcome, a subgroup cell in place of the whole group), 20 dispersions of the wrong type (a standard error or an interval width written as a standard deviation), 6 arm swaps, 2 sign inversions (survivors in place of deaths), 2 field mixes and 17 values the text does not print — eight ratio-shaped strings and one count reconstructed from a percentage and an inconsistently printed n , all from `llama3.1:8b` on anchor 1, and eight on anchor 2, from four models (three change means, four dispersions and one denominator); none reached anchor 3. Twenty-one cells were errors of the grading program itself, not of the model: the model’s answer was right and came to count as a hit in the scores; the models’ answers and the programs were not altered, and in the pooled estimates the effect appears only as a sensitivity analysis. They were ten copies of one same case in which the reversal lens replaced a fragment inside a correctly transcribed number, one altered value written with a minus sign the lens does not recognize, and ten cells in which a model applied the key’s own derivation rule (change = final – baseline) to the altered text and landed on a value the lens had never seen. Thirty-nine cells put the answer key itself on trial — three trials that print both the randomized and the analyzed count under a field called *n randomized*, where the key holds the one the review pooled — and were listed for the author without being flipped.

Table 7: Verdicts on the 619 discordant cells, by anchor.

Verdict	Anchor 1	Anchor 2	Anchor 3	All
Faithful to the source, other encoding or layer	197	149	0	346
Omission	65	26	56	147
Reading slip	25	73	7	105
Model correct, grader wrong	10	11	0	21
Non-matching cells	297	259	63	619

4 Discussion

Six small language models, running on a consumer computer and reading altered copies of 29 randomized trials, rebuilt two published meta-analyses at their published estimates and the third as its trials support it, not as published. Under the alteration control none of this is attributable to memory of the reviews, and under the trial-verified answer key the published reviews were graded on the same ruler as the models.

The decisive clinical result is that of the third meta-analysis. Its published conclusion — current randomized evidence does not demonstrate a mortality benefit of methylene blue in shock — depends on two transcription errors: survivors entered as deaths in one trial and the deaths of two arms summed over the denominator of one in another. With the data as the

trials publish them, the odds ratio of death falls from 0.73 [0.40; 1.36] to 0.48 [0.34; 0.69], and neither correction taken alone, nor the exclusion of the disputed trial, restores the published conclusion (Table 2). The corrected estimate is not a new trial: they are the same eight small trials, the trial sequential analysis reported beside the published result was not recomputed, and the review’s own Table 1 already recorded a mortality reduction for the inverted trial. No sheet of any model reproduced the errors, four models read the Spanish-language trial as it prints, all six enumerated the three-arm trial and two arrived at the corrected estimate to the third decimal place. Three things bound this result, and the protocol wrote them before the first call: the errors were raised by this study’s adjudicator while building the answer key, so the models were measured against them, not credited for finding them; the anchor-3 sheet asks four fields per arm over enumerated arms, the two axes on which the human extractors failed, so the instrument helps a reader get the correction right; and the event counts carry no reading proof — it rests on altered denominators in four and a half trials of eight — while the trials, unlike the reviews, predate the models’ training, so a recited death count cannot be excluded, although the review’s recited cell that would betray memory of the meta-analysis never appeared. What the anchor establishes is narrower and useful: a trial-faithful local reader produces the corrected meta-analysis with no instruction at all to look for errors, and a test grading against the published cells would have given zero to its two best readers on the two cells that matter.

Human errors and model errors have different profiles and one point in common. That the human column carries errors is consistent with the methodological literature, and not remarkable within it [Gøtzsche et al., 2007, Jones et al., 2005, Tendal et al., 2011, Mathes et al., 2017]; what this study adds is the cell-by-cell quotation trail and an audited model column beside the human one, on the same ruler: 102 of 124, 49 of 49 and 29 of 32 cells for the reviews, against 85 to 111, 16 to 36 and 16 to 32 for the models. The confirmed human errors are transcription and labelling; the models’ concentrate in omissions and in counting another population. The conclusion-weight errors on either side — survivors entered as deaths, a numerator summed over one arm’s denominator, a dispersion of the wrong type — are wrong labels, which suggests that the semantics of the labels, not the numbers, is where extraction fails for both kinds of extractor and where the checking of a meta-analysis should concentrate. The choice of the population counted, the largest faithful class among the models’ discordances, is the same ambiguity human reviewers face: in one trial, every sheet counted the patients who completed the study, which the article puts forward, while the review had pooled all those randomized (intention-to-treat analysis), and both counts are printed.

For whoever assembles or checks a meta-analysis, the split of difficulty between the outcomes is the practical finding. Dichotomous material is transcription — counts and denominators printed redundantly in table, prose and percentage — and the calculation chain dampens the error that remains, because a risk ratio shifts numerator and denominator together when another population is counted: twelve sheets took visibly different denominator routes and still landed within 0.05 of the verified morbidity estimate. Each continuous quantity is printed once, in one of several competing versions, and the decisive step is semantic — identifying what a printed dispersion *is*; a mislabelled type sends the conversion down the wrong route, and pooling amplifies the error, because it weights each trial by the inverse of the variance [Higgins et al., 2023]. The dispersion-type slips that concentrated on the continuous anchor are the class that an audit of one published meta-analysis found in 9 of 16 human-extracted effect sizes, where correcting the standard-error/standard-deviation mix-up moved the pooled estimate from 1.46 to 0.54 [Sandercock, 2024]; that class deserves a dedicated check, sheet against trial, more than a better reader. None of the reading failures was fabrication: every divergent cell traces back to the text read or to a calculation on it, which is consistent with a cloud-model extraction study in which *omitted data were the most common error type* [Gartlehner et al., 2024]. Dichotomous outcomes can therefore be delegated broadly to a local reader; continuous outcomes concentrate

the risk and call for the best reader and the strongest check.

The computational part can be summarized in a few sentences. Everything ran on a single consumer mini-computer, sending no text to any service, which meets the confidentiality condition of clinical text and the rigor and transparency requirements of the evidence-synthesis organizations [Fleming et al., 2025]. The ranking of the models depended on the outcome: **qwen3.8:27b** (27 billion parameters) read best on the two anchors where competing versions of the same datum exist, and two 14-billion models (**qwen3:14b** and **deepseek-r1:14b**) on the count anchor — an observation, not a tested hypothesis. The mandatory-quotation sheet cost accuracy on the descriptive anchor, where the trial prints table rows rather than sentences; on the continuous anchor, after the grader correction, it scored the same or higher and brought four of six estimates closer to the trials, one anchor short of the pre-registered criterion, so it is not adopted as the default. Its value — a reviewer can check a cell without opening the article — is a usage decision. The 21 right answers that the graders themselves had marked wrong came to count as hits by the rule of quoting before judging, which is also the constraint on plausibility bias, the tendency of any judge, human or model, to prefer the plausible reading to the printed one [Zheng et al., 2023].

The next study, already designed, is a v3 sheet in which the model only transcribes what is printed, each value with its quotation and its type label (standard deviation, standard error, confidence interval), and the grading program does the derivations by the same rules as the answer key; the quotation may be the table row, and the text converted from PDFs is cleaned of invalid characters before reading. The hypotheses, fixed before any run, are to keep a quotation for every printed value without losing completeness and to reduce wrong-type dispersions to near zero, the largest group of reading errors in this study. The protocol is in the repository (Study 13), to be registered before the first call.

Three design choices on anchor 3, imposed by the review’s inconsistencies described in Section 2.3, were proposed by the assistant, submitted to the author before any run and kept; they are made explicit here, with the alternative and the cost of each, so that a reader can disagree with them without having to reconstruct them from the data. The reading proof alters denominators with coherent recalculation of the percentages, rather than altering only percentages, which would make count, denominator and percentage contradict one another within the trial, or leaving the anchor unaltered; it buys a partial proof whose limits accompany every anchor-3 number. The forest plot governs among the three incompatible descriptions the review gives of its primary analysis because it is the only one with cells attached. Aguilar’s 21-day value is pooled in an analysis the review labels 28–30 days because that is what the review did, with its exclusion run as a recorded sensitivity: 0.484 with the trial, 0.493 without it, the conclusion unchanged either way.

Several limitations bound these claims. Three meta-analyses and 29 trials are the entire evidence base, two of the reviews from the same journal family; every superlative is restricted to these corpora, and no inference across models is attempted. One compression per model and one hardware profile were tested, so size and precision are not separable. On anchor 3 the reading proof covers the denominators of four and a half trials and no event count, the trials predate the models’ training, the errors were raised by the adjudicator before any model read the review, and the sheet is built on the axes where the human extractors failed. The adjudication of the 619 cells was delegated by the author to the AI assistant under the rule of quoting before judging, which trades independence for completeness; the independent re-reading that reversed nine of the reader’s 85 verdicts mitigates but does not remove this. The human rows score the published cells against a key built by verifying those same cells, so their divergences are, by construction, the verification record itself. The automatic comparator approximates the adjudication ruler, so the absolute scores underestimate agreement. Finally, the gaps of the alteration program — percentages, totals and repeated sentences that survive — are documented and exploitable, although what they produced here was reconstruction, not

recitation.

In conclusion, a consumer computer running a well-chosen local reader, with frozen pooling programs, rebuilt three published meta-analyses from their trials as the trials support them: two at their published estimates, and one at a corrected estimate whose clinical conclusion differs from the published one. The next question is how often checking against the trials, rather than against the review, would change published estimates if applied at scale.

Data and code availability

The pre-registered protocol with its review, the frozen instruments in English, the alteration maps with SHA-256 fingerprints, the two-layer answer keys with their trial-verification records, the anchor-3 errata record and the held alert table, the pooling and grading programs, the campaign execution program, all 696 raw model outputs with per-call records, the Phase-2 and Phase-4 records with the 619 adjudicated cells and their quotations, the review of the reader’s verdicts and the exact compressed models, pinned by Ollama identifier, are public at <https://github.com/gbbarra/llm-evidence-extraction-benchmark-ptbr> and archived on Zenodo [Barra, 2026]. Paywalled trials are not redistributed; scripts rebuild that stratum of the corpus from legally obtained copies.

Declarations

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Conflicts of interest. The author declares no conflicts of interest. The AI assistant declared below was used under an ordinary consumer subscription paid by the author, who has no financial or professional relationship with Anthropic or with the organizations whose open-weight models are evaluated here.

Use of artificial intelligence. The study was designed and conducted by the author with the assistance of Claude (Anthropic), which drafted the protocol from the author’s design decisions (three assistant-originated choices were submitted to the author before any run), built the anchor-3 answer key, alteration instrument and execution program, implemented the graders and the adjudication rules, performed the Phase-4 adjudication under delegation from the author with every verdict tied to a quotation and reversible, and drafted this manuscript under the author’s direction and review. The local models evaluated produced only the extraction sheets graded here; every downstream number was produced by frozen code and verified mechanically. In line with the position of the evidence-synthesis organizations [Flemyng et al., 2025], every AI contribution is reported and the author retains responsibility for the work.

Ethics approval and consent. Not applicable: no human participants and no new data collection; only published aggregate reports were analyzed.

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